LCIA Reference — Life Cycle Impact Assessment

What is LCIA?

LCIA stands for **Life Cycle Impact Assessment**. It is the third phase of an LCA (after Goal & Scope and Inventory Analysis). The job of LCIA is to take the **inventory table** – the list of all emissions and resource extractions from the supply chain — and translate those numbers into **meaningful environmental impacts**.

Think of it like a restaurant menu's nutritional information panel:

- The **inventory** (LCI) tells you "this meal contains 15 g of saturated fat, 120 mg of sodium, and 8 g of sugar."
- The **impact assessment** (LCIA) tells you "this meal contributes an estimated X% towards your daily risk of heart disease, Y% towards high blood pressure, and Z% towards diabetes risk."

Same idea — just for the planet instead of a person.

The LCIA Framework

LCIA is built from four sequential steps:

```
Inventory (LCI) → Classification → Characterization → Normalisation & Weighting
                                            |
                                    [Midpoint level]
                                            |
                                 Damage modelling
                                            |
                                    [Endpoint level]
```

The two most important boxes are **Classification** and **Characterization**. Normalisation and weighting are optional (and less used in teaching LCA).

Step 1 — Classification

Classification is the act of assigning each emission or extraction from the inventory to one or more **impact categories**.

An *impact category* is a type of environmental problem the emission contributes to. Common examples:

Impact category	What it measures	Example contributor
Climate change (global warming)	Heat trapped in the atmosphere	CO_2, CH_4, N_2O
Eutrophication	Excess nutrients in water bodies	NO_x, NH_3, phosphates
Acidification	Acid rain (damages soil, water, buildings)	SO_2, NO_x
Ecotoxicity	Harm to plants and animals	Heavy metals, pesticides
Human toxicity	Harm to human health	Particulate matter, benzene
Ozone depletion	Thinning of the stratospheric ozone layer	CFCs, halons
Resource depletion (abiotic)	Using up non-renewable resources	Crude oil, copper ore

Some substances contribute to *multiple* categories. For example, NO_x contributes to both **climate change** and **acidification** and **eutrophication**. Classification simply notes that NO_x belongs to all three.

The classification step is trivial once you have a method — you just look up each emission in a table to see which categories it affects.

Step 2 — Characterization

Characterization is where the actual maths happens. For each substance in each category, we multiply its inventory amount by a **characterization factor** (CF) to convert it into a **common unit** for that category.

Why is this needed?

Different emissions in the same category have different *potencies*. For climate change:

- 1 kg of CO_2 has a global warming potential (GWP) of **1** (it is the reference)
- 1 kg of CH_4 has a GWP of **28** (it traps 28x more heat over 100 years)

- 1 kg of N_2O has a GWP of **265**

So you cannot just add kilograms of each gas together. You must characterise each one first.

The characterization formula

$$I_c = \sum_i (m_i \times CF_{i,c})$$

Where:

Symbol	Meaning
I_c	Total impact score for category c
m_i	Mass (in kg) of substance i emitted
$CF_{i,c}$	Characterisation factor for substance i in category c

Example — climate change from a coffee cup

Substance	Amount (kg)	CF (kg CO_2-eq / kg)	Contribution
CO_2	0.10	1	0.10 kg CO_2-eq
CH_4	0.002	28	0.056 kg CO_2-eq
N_2O	0.0005	265	0.1325 kg CO_2-eq

$$I_{\text{climate}} = 0.10 + 0.056 + 0.1325 = 0.29 \text{ kg CO}_2\text{-eq}$$

So the cup's total contribution to climate change is **0.29 kg of CO_2-equivalent** – a standardised unit that lets you compare across different greenhouse gases.

This same logic repeats for every impact category. For acidification, the common unit is **kg SO_2-eq**. For eutrophication, it is **kg $PO_4$3–eq**. For resource depletion, it is **kg Sb-eq** (antimony equivalents).

Step 3 — Midpoints vs Endpoints

This is the most important conceptual distinction in LCIA.

Midpoint indicators

A **midpoint** is a point *along the cause-effect chain*, before the final harm. It answers the question "how much does this contribute to a specific environmental problem?".

Examples of midpoint indicators:

Midpoint category	Unit	What it tracks
Climate change	kg CO_2-eq	Heat trapped
Acidification	kg SO_2-eq	H+ ions released
Eutrophication	kg $PO_4$3–eq	Nutrients released
Ecotoxicity	CTUe	Potentially affected fraction of species
Ozone depletion	kg CFC-11-eq	Ozone destroyed

Midpoints are relatively certain (the science is well-established for most of them) and are the standard output of most LCA studies.

Endpoint indicators

An **endpoint** is the *final harm* to something we care about. Endpoints go further along the cause-effect chain to ask "so what does this actually damage?".

The three endpoint Areas of Protection (AoP) in most LCIA methods are:

Area of Protection	What it measures	Unit
Human health	Years of healthy life lost	DALY (Disability-Adjusted Life Years)
Ecosystem quality	Species lost over time	species-yr
Resource availability	Extra cost to extract future resources	USD

Going from a midpoint to an endpoint requires **damage modelling** – and that introduces more uncertainty.

The pathway from midpoint to endpoint

Take climate change as an example:

```
Climate change midpoint (kg CO₂-eq)
        v
Temperature increase (°C)
        v
Crop yield changes, disease spread, sea level rise
        v
Human health: DALYs from malnutrition, malaria, flooding
Ecosystem quality: species lost from habitat change
Resource availability: agricultural productivity loss
```

Each arrow is a *damage model* – a scientific estimate of how a change at one level propagates to the next. Different LCIA methods use different damage models, which is why you can get slightly different endpoint results from the same inventory.

The math

$$\text{Endpoint}_k = \sum_c (I_c \times DF_{c,k})$$

Where:

Symbol	Meaning
Endpoint_k	Damage score for Area of Protection k (e.g. human health in DALY)
I_c	Midpoint impact score for category c (e.g. kg CO_2-eq)
$DF_{c,k}$	Damage factor linking midpoint c to endpoint k

Example — climate change midpoint to endpoint

Assuming a damage factor of 2.4×10^{-6} DALY per kg CO_2-eq (a typical ReCiPe value):

$$\begin{aligned}\text{Endpoint}_{\text{human health}} &= 0.29 \text{ kg CO}_2\text{-eq} \times 2.4 \times 10^{-6} \text{ DALY/kg CO}_2\text{-eq} \\ &= 7.0 \times 10^{-7} \text{ DALY}\end{aligned}$$

This means the coffee cup’s contribution to global warming causes an estimated loss of **0.7 millionths of a healthy life-year** – a very small number for one cup, but significant when scaled to millions of cups per year.

Visual summary of the chain

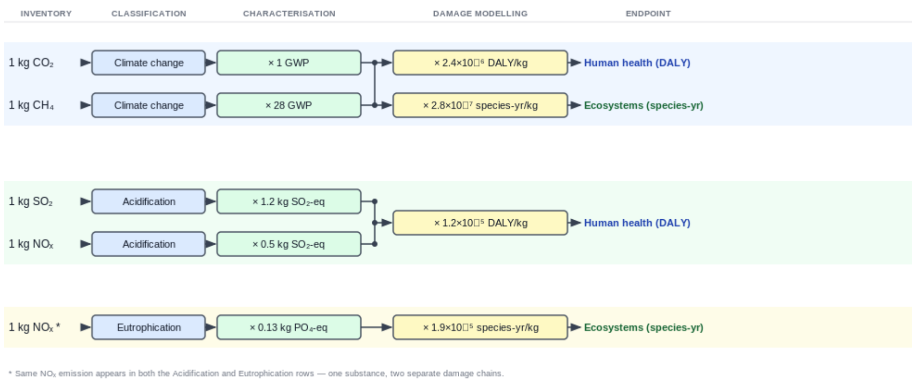

Figure 1: LCIA chain diagram

The three levels of indicator certainty

Level	Certainty	Meaning for decision-making
Midpoint	High	Scientifically robust. Compare products by kg CO_2-eq, kg SO_2-eq, etc.
Endpoint (single score)	Lower	The damage modelling introduces assumptions. Useful for communicating with non-specialists.
Weighted single score	Lowest	Requires value judgements (is climate change more important than toxicity?). Depends on whose values you use.

Most LCA studies report midpoints. Endpoints are often added for communication. Weighted single scores are used when a single-number answer is needed (e.g. “Product A is 30% greener than Product B”).

Common LCIA methods

Method	Origin	Midpoints	Endpoints	Notes
ReCiPe 2016	Netherlands	18	3 (human health, ecosystems, resources)	Most widely used. Hierarchist perspective is the default.
CML 2002 / 2016	Netherlands	10	None (midpoint only)	Classic academic method.
TRACI 2.1	USA (EPA)	10	None	Common in North American LCA.
ILCD 2011	EU Commission	16	None	Recommended by the European Commission.
IMPACT World+	International	18	4 (adds water and carbon)	Good for water footprinting.
EF 3.0 / 3.1	EU Commission	16	4	Used for EU Product Environmental Footprint (PEF).

Worked hypothetical example – “WidgetCo” desk light

Let’s walk through a full LCIA from inventory to midpoint to endpoint for a fictional product.

Inventory (from LCI)

Substance	Amount
CO_2 to air	5.0 kg
CH_4 to air	0.05 kg
SO_2 to air	0.02 kg
NO_x to air	0.01 kg
Phosphate to water	0.001 kg

Step 1 — Classification

Substance	Affected categories
CO_2	Climate change
CH_4	Climate change
SO_2	Acidification
NO_x	Acidification, Eutrophication
Phosphate	Eutrophication

Step 2 — Characterisation (midpoints)

We use ReCiPe 2016 (Hierarchist) midpoint factors:

Substance	Amount	Factor	Midpoint score	Unit
CO_2	5.0 kg	$\times 1$	$=$ **5.0**	kg CO_2-eq
CH_4	0.05 kg	$\times 28$	$=$ **1.4**	kg CO_2-eq
			6.4 kg CO_2-eq	*(Total climate change)*
SO_2	0.02 kg	$\times 1.0$	$=$ **0.02**	kg SO_2-eq
NO_x	0.01 kg	$\times 0.5$	$=$ **0.005**	kg SO_2-eq
			0.025 kg SO_2-eq	*(Total acidification)*
NO_x	0.01 kg	$\times 0.13$	$=$ **0.0013**	kg PO_4-eq
Phosphate	0.001 kg	$\times 1.0$	$=$ **0.001**	kg PO_4-eq
			0.0023 kg PO_4-eq	*(Total eutrophication)*

Step 3 — Midpoint → Endpoint (damage modelling)

Using ReCiPe 2016 damage factors (Hierarchist):

Midpoint score	× Damage factor	= Endpoint contribution
6.4 kg CO_2-eq	$\times 2.4 \times 10^{-6}$ DALY/kg	$= 1.5 \times 10^{-5}$ DALY (human health)
6.4 kg CO_2-eq	$\times 2.8 \times 10^{-7}$ species-yr/kg	$= 1.8 \times 10^{-6}$ species-yr (ecosystems)
0.025 kg SO_2-eq	$\times 1.2 \times 10^{-5}$ DALY/kg	$= 3.0 \times 10^{-7}$ DALY (human health)
0.0023 kg PO_4-eq	$\times 1.9 \times 10^{-5}$ species-yr/kg	$= 4.4 \times 10^{-8}$ species-yr (ecosystems)

Summing damage contributions:

$$\begin{aligned}
\text{Human health} &= 1.5 \times 10^{-5} + 3.0 \times 10^{-7} \\
&= 1.5 \times 10^{-5} \text{ DALY} \\
\text{Ecosystem quality} &= 1.8 \times 10^{-6} + 4.4 \times 10^{-8} \\
&= 1.8 \times 10^{-6} \text{ species-yr}
\end{aligned}$$

The climate change midpoint dominates both endpoint categories — it accounts for nearly all the damage in this example.

Key terms summary

Term	Definition
LCIA	Life Cycle Impact Assessment — turning inventory data into environmental impact scores
Impact category	A type of environmental problem (climate change, acidification, etc.)
Classification	Assigning each emission to the categories it affects
Characterisation	Multiplying by characterisation factors to get a common-unit score

Term	Definition
Characterisation factor (CF)	The potency of a substance in a given category (e.g. GWP = 28 for CH_4)
Midpoint	An indicator along the cause-effect chain, before final damage (kg CO_2-eq)
Endpoint	An indicator of final damage to something we value (DALY, species-yr)
Damage factor (DF)	The conversion factor from a midpoint to an endpoint
Normalisation	Dividing by a reference value (e.g. total EU emissions) to see relative significance
Weighting	Applying value-based weights to combine categories into a single score
Area of Protection (AoP)	The three things we ultimately want to protect: human health, ecosystems, resources
DALY	Disability-Adjusted Life Year — a measure of overall disease burden (1 DALY = one lost year of healthy life)

Further reading

- **ReCiPe 2016** – www.rivm.nl/en/life-cycle-assessment-lca/recipe
- **ILCD Handbook** – European Commission Joint Research Centre
- **ISO 14044** – Section 4.4 (LCIA requirements in the ISO standard)
- **Jolliet et al. 2016** – *Environmental Life Cycle Assessment*, Chapter 5